

Indian Statistical Institute

Course Name: M. Math, 2nd year

Subject Name : Topology III

Exam name : HW 1

Due: August 14, 2025

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- You may use any results proved in class. Any other results (including those in homework problem sets) require proof. You are required to provide proper mathematical reasons for your answers.

1. Let X be an n -manifold, such that $S(X)$, the unreduced suspension of X , is an $(n+1)$ -manifold. Prove that $\tilde{H}_*(X) \cong \tilde{H}_*(S^n)$.
2. Let $\eta : S^3 \rightarrow S^2$ be the Hopf map, that is, the attaching map for $\mathbb{C}P^2$. Consider the map τ expressed as the composite

$$\tau : S^3 \rightarrow S^3 \vee S^3 \xrightarrow{\eta \vee \eta} S^2 \vee S^2,$$

where the map $S^3 \rightarrow S^3 \vee S^3$ is the map obtained by pinching the equatorial S^2 to a point. Let P be the mapping cone of τ .

- a) Let $r_1 : S^2 \vee S^2 \rightarrow S^2$ be the map which quotients out the second factor. Prove that $r_1 \circ q$ is homotopic to η . Using this produce a map $P \rightarrow \mathbb{C}P^2$ which induces the identity map on H^4 .
 - b) By constructing maps similar to the one in a), compute the ring structure on $H^*(P)$.
 - c) Is $P \simeq S^2 \times S^2$?
3. Compute the ring $H^*(X; \mathbb{Z})$ when X is obtained from $\mathbb{C}P^{n-1} \vee S^{2n-1}$ by attaching a cell e^{2n} by the composition

$$S^{2n-1} \rightarrow S^{2n-1} \vee S^{2n-1} \xrightarrow{f \vee g} S^{2n-1} \vee \mathbb{C}P^{n-1}$$

where the first map collapses the equator $S^{2n-2} \subset S^{2n-1}$ to a point, the map f has degree $k > 1$, and g is the attaching map for the $2n$ -cell of $\mathbb{C}P^n$.